6B Graphing Linear Relations and Modelling

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. State the gradient m and y-intercept c of each linear relation:

(a) $y = 5x - 8$ (b) $y = -3x + 11$ (c) $y = 9 - 2x$ (d) $y = \frac{3}{4}x$

Q2. Calculate the gradient m of the line passing through each pair of points:

(a) $(1, 4)$ and $(5, 16)$ (b) $(0, 10)$ and $(4, -2)$ (c) $(-3, -5)$ and $(2, 15)$ (d) $(-4, 9)$ and $(6, 4)$

Q3. Find the coordinates of the y-intercept and the x-intercept for each line:

(a) $y = 3x - 12$ (b) $y = -2x + 10$ (c) $4x + 5y = 20$ (d) $3x - 2y = 18$

Q4. Write the equation of each straight line in gradient-intercept form $y = mx + c$:

(a) Gradient 4, y-intercept -7 (b) Passes through $(0, 5)$ and $(3, 17)$ (c) Passes through $(0, -3)$ and $(4, -11)$ (d) Passes through $(2, 7)$ and $(5, 16)$

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. A hot-water urn at a community footy club starts at 20°C and heats up at a constant rate of 8°C per minute. (a) Write a linear rule for the temperature T\text{ (^\circC)} after t minutes. (b) How many minutes does it take to reach boiling point (100°C)?

Q6. A prepaid mobile data pack starts with $45\ \text{GB}$ of data and uses $1.5\ \text{GB}$ per day. (a) Write a rule for the remaining data D (GB) after t days. (b) How much data remains after 18 days? (c) After how many days is the data completely used up?

Q7. A delivery van leaves Bendigo ($150\ \text{km}$ from Melbourne) and drives towards Melbourne at a constant speed of $75\ \text{km/h}$. Write a linear rule for its distance d (km) from Melbourne after t hours, and state the d-intercept and t-intercept with their physical meanings.

Q8. A straight line has equation $2x + 3y = 18$. Find its x- and y-intercepts and calculate the area of the right-angled triangle enclosed by the line and the two coordinate axes.

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. Two lines have equations $y = 3x - 4$ and $y = 3x + 5$. Explain using their gradients and y-intercepts why these two lines are parallel and never intersect.

Q10. Determine whether the three points $P(-1, -5)$, $Q(2, 4)$, and $R(6, 16)$ lie on the same straight line (are collinear) by comparing the gradient of PQ with the gradient of QR.